

Estimation Theory

Lecture Notes for Engineering (Computer Science) Students

Learning Objectives

After this lecture, students will be able to:

- Understand the concept of estimation.
- Explain point estimation and interval estimation.
- Understand properties of good estimators.
- Apply estimation techniques in computer science.
- Understand confidence intervals.
- Appreciate the role of estimation in Machine Learning, Artificial Intelligence, Big Data, and Software Engineering.

Motivation:

Start with a Question: Suppose Google wants to know

"What is the average time users spent on YouTube daily?"

Can Google ask all 2.5 billion users? **No.**

Instead,

- Select a sample
- Calculate the average
- Estimate the population average

This process is called **Estimation**.

Another Example: Suppose Amazon wants to know Average delivery time across India.

Instead of checking millions of deliveries, they inspect 20,000 deliveries and estimate the average delivery time.

Definition of Estimation Theory:

Estimation Theory is a branch of statistics that deals with estimating unknown population parameters using information obtained from a sample.

Since studying an entire population is often expensive or impractical, a sample is selected, and its statistics are used to estimate the corresponding population parameters.

Example:

A software company wants to estimate the average daily internet usage of all its users.

Total users = 5 million (population)

Randomly selected users = 10,000 (sample)

Average internet usage of the sample ($\bar{x}$) = 4.2 GB/day

Therefore, the company estimates that the average internet usage of all users is approximately 4.2 GB/day. i.e., $\mu = 4.2 \text{ GB/day}$

Why is estimation needed?

In real life, population parameters are almost always unknown.

Examples:

- 1) Average salary of software engineers in India
- 2) Average CPU temperature
- 3) Average internet speed
- 4) Average rainfall
- 5) Average battery life, we estimate these values.

Real-life Computer Science Example: Suppose Microsoft tests 500 laptops

Average battery backup = 7.8 hours

Instead of testing 5 million laptops, they estimate Population Mean ≈ 7.8 hours

Types of Estimation

There are two types. 1. **Point Estimation**, 2. **Interval Estimation**

What is a Point Estimator?

Definition:

A point estimator is a statistic calculated from a sample that provides a single numerical value as an estimate of an unknown population parameter.

For example: Sample mean ($\bar{x}$) estimates the population mean (μ).

Sample proportion (p) estimates the population proportion (p).

Example:

A teacher wants to estimate the average marks of 500 students.

He randomly selects 50 students and obtains the following average: $\bar{x} = 72$

The sample mean 72 is the point estimate of the population mean.

What is an Interval Estimation?

Definition:

Interval estimation is a statistical method used to estimate an unknown population parameter by providing a range of values (called a confidence interval) instead of a single value.

This range is expected to contain the true population parameter with a specified confidence level (such as 90%, 95%, or 99%).

In other words,

Interval estimation provides a lower limit and an upper limit within which the true population parameter is likely to lie.

Unlike point estimation, which gives only one estimate, interval estimation indicates the uncertainty associated with the estimate.

Example:

Teacher asks Average marks?

Point Estimate is 72

Interval Estimate is 68–76

Which is better?

Interval estimation. Because uncertainty is included.

Estimator:

The Procedure or rule to determine an unknown population parameter is called an estimator

Point Estimator is a statistic used to estimate a parameter.

Examples:

Population Mean Estimated using Sample Mean

Population Variance Estimated using Sample Variance

Population Proportion Estimated using Sample Proportion

Common Point Estimators

Population Parameter	Point Estimator
Mean μ	Sample Mean $\bar{x}$
Variance σ^2	Sample Variance S^2
Proportion P	Sample Proportion $\hat{p}$

Example

Marks: 70, 75, 80, 85, 90

Average: $= (70+75+80+85+90)/5 = 80$

Population Mean Estimated as 80

Distinguish Between Point Estimation and Interval Estimation.

Point Estimation	Interval Estimation
Provides a single value to estimate a population parameter.	Provides a range of values within which the parameter is expected to lie.
Does not indicate uncertainty.	Indicates the uncertainty of the estimate through a confidence level.
Easier to calculate.	Requires standard error and a confidence level.
Used for quick estimation.	Used for more reliable statistical inference.

Characteristics of Good Estimators

A good estimator should satisfy four properties.

1) Unbiasedness:

Definition: An estimator is unbiased if its expected value equals the true parameter.

Expected value equals true parameter. i.e., $E(\hat{\theta}) = \theta$

Mathematically, $E(\hat{\theta}) = \theta$, where, $\hat{\theta} = \text{Estimator}$

$\theta = \text{Population Parameter}$

If $E(\hat{\theta}) \neq \theta$ the estimator is biased, and the bias is calculated as

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

Example:

- 1) Actual average weight 70 kg.

Repeated samples give ,69.9, 70.2, 70.1, 69.8

Average: 70

Estimator is unbiased.

- 2) If the true average height is **170 cm** and repeated random samples produce averages that centre around **170 cm**, then the estimator is unbiased.

2) Consistency:

A consistent estimator becomes closer to the true parameter as sample size increases.

Mathematically, $\hat{\theta} \rightarrow \theta$, as $n \rightarrow \infty$

i.e., As sample size increases, estimate approaches true value.

Example:

Sample size $n = 20$ Average height is 165 cm

Sample size $n = 500$ Average height is 168 cm

Sample size $n = 5000$ Average height is 168.3 cm

Closer to actual value.

3) Efficiency:

Among all unbiased estimators, the one having the smallest variance is called efficient.

Efficient estimators provide more stable estimates.

If $\hat{\theta}_1$ is the most efficient estimator with Variance V_1 , and $\hat{\theta}_2$ is any other estimator with Variance V_2 , then the efficiency E is defined as $E = (V_1 / V_2) \leq 1$, i. e. , $V(\theta_1) < V(\theta_2)$

In otherwords,

If two of the Consistent estimators θ_1, θ_2 of a Certain parameter θ , we have $V(\theta_1) < V(\theta_2)$, for all n , then θ_1 is more efficient than θ_2 for all sample sizes.

4) Sufficiency:

An estimator is sufficient if it contains all the information in the sample regarding the parameter.

For normally distributed data, Sample Mean is sufficient for estimating the population mean.

Example: Sample Mean uses every observation.

Hence, Sample Mean is sufficient for estimating μ (under common statistical models).

Interval Estimation and Confidence Intervals

A **confidence interval (CI)** provides a range of plausible values for an unknown population parameter. For a population mean with **known population standard deviation**,

$$\bar{x} \pm z_{(\alpha/2)} \left(\frac{\sigma}{\sqrt{n}} \right)$$

where:

$\bar{x}$ = sample mean

σ = population standard deviation

n = sample size

$Z_{\alpha/2}$ = critical value from the standard normal distribution

For 95% confidence, $Z_{\alpha/2} = 1.96$,

If σ is unknown, replace Z with the appropriate t-value and σ with s .

Example:

A random sample of 100 students has

- Sample mean = 72 marks
- Population standard deviation = 10 marks

The standard error is $SE = 10/100 = 1$

Hence, the 95% confidence interval is $72 \pm 1.96 (1) = 72 \pm 1.96$

or

$$(70.04, 73.96)$$

Interpretation:

We are 95% confident that the true average marks of all students lie between 70.04 and 73.96.

This is confidence interval.

Formula for Known variance: $\bar{x} \pm z_{(\alpha/2)} \left(\frac{\sigma}{\sqrt{n}} \right)$

Formula for Unknown variance: $\bar{x} \pm t_{(\alpha/2)} \left(\frac{s}{\sqrt{n}} \right)$

The computation of confidence interval and confidence limits based on the sampling distribution of a static. By central limit theorem, we know that the sampling distribution of the sample means $\bar{x}$ of a random sample of size n ($n \geq 30$) drawn from a population having mean μ and standard deviation σ is approximately a normal distribution with mean μ and S.D = S. E. of $(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

$\therefore Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ has a standard normal distribution with mean = 0 and S. D. = 1

Confidence Limits for population mean μ at 95 % confidence is given by $|Z| \leq 1.96$

$$\text{i.e., } \left| \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \right| \leq 1.96$$

$$|\bar{x} - \mu| \leq 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$-1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \leq \bar{x} - \mu \leq 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

Multiplying with (-1) on both sides

$$1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \geq -\bar{x} + \mu \geq -1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

Add $\bar{X}$ on both sides, then we get

$$\bar{X} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \geq \mu \geq \bar{X} - 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \quad \text{rewrite the inequality as}$$

$$\bar{X} - 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) \quad \text{these are called Confidence}$$

Limits for population mean μ at 95 % confidence.

i.e., $\bar{x} \pm z_{(\alpha/2)} \left(\frac{\sigma}{\sqrt{n}} \right)$ are Confidence limits for μ

$z_{(\alpha/2)} \left(\frac{\sigma}{\sqrt{n}} \right)$ is called maximum error of estimate (E)

Similarly, 99 % confidence Limits for population mean μ is given by

$$\bar{X} - 2.58 \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + 2.58 \left(\frac{\sigma}{\sqrt{n}} \right)$$

Similarly, 90 % confidence Limits for population mean μ is given by

$$\bar{X} - 1.64 \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + 1.64 \left(\frac{\sigma}{\sqrt{n}} \right)$$

Confidence Limits for population Proportion (P)

1. 95 % confidence Limits for population proportion is given by

$$p - 1.96 \left(\sqrt{\frac{PQ}{n}} \right) \leq P \leq p + 1.96 \left(\sqrt{\frac{PQ}{n}} \right)$$

2. 99 % confidence Limits for population proportion is given by

$$p - 2.58 \left(\sqrt{\frac{PQ}{n}} \right) \leq P \leq p + 2.58 \left(\sqrt{\frac{PQ}{n}} \right)$$

3. 90 % confidence Limits for population proportion is given by

$$p - 1.64 \left(\sqrt{\frac{PQ}{n}} \right) \leq P \leq p + 1.64 \left(\sqrt{\frac{PQ}{n}} \right)$$

Real-Life Applications in Computer Science

Application 1 - Machine Learning:

Estimate model accuracy

Instead of evaluating on every possible future input, estimate accuracy using Test Dataset.

Example: Accuracy 95% Confidence Interval 94–96%

Application 2 - Artificial Intelligence:

Estimate Probability of spam email.

Estimated probability 0.92 Model predicts Spam.

Application 3 - Cloud Computing:

Estimate Average server load.

AWS continuously estimates CPU utilization Memory usage

Network traffic.

Application 4 - Software Engineering:

Estimate Bug density.

Example: 200 bugs in 100000 lines Estimate bugs per KLOC.

Application 5 - Cyber Security:

Estimate Probability of attack. Estimate Average login failures.

Estimate Fraud detection rate.

Application 6 - Recommendation Systems:

- Netflix estimates movie ratings.
- Amazon estimates Product preference.
- Spotify estimates Music preference.

Application 7 - Social Media:

Instagram estimates average engagement.

Facebook estimates advertisement click rate.

Classroom Example

Suppose 20 students CGPA are 7.2, 7.8, 8.1, 8.4, 8.0,, 7.9

Average is 7.88

Estimated average CGPA Entire class ≈ 7.88